Making Safe Reinforcement Learning Faster

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Situation

Configurable Environments

Most Reinforcement Learning research focuses on **static MDPs**, where the environment dynamics are fixed.

However, many real-world environments are **configurable**—certain parameters can be tuned to improve agent performance.

Configurable MDPs (Conf-MDPs) extend standard MDPs by allowing the transition model P to be selected from a space $\mathcal{P}$:

$$\mathcal{CM} = (\mathcal{S}, \mathcal{A}, R, \gamma, \mu, \mathcal{P}, \Pi)$$

Example

Car Configuration

Consider a car with **sport** and **comfort** driving modes:

- **Sport mode:** Aggressive acceleration, less stability
- **Comfort mode:** Smooth driving, better control

On a **slippery road**, the optimal configuration depends on the driving policy—and vice versa.

The goal: **jointly optimize** the policy π and environment configuration P to maximize expected return $J_\mu^{P,\pi}$.

Problem & Solution

The Challenge

SPMI (Safe Policy-Model Iteration) provides **monotonic improvement guarantees** but is computationally expensive.

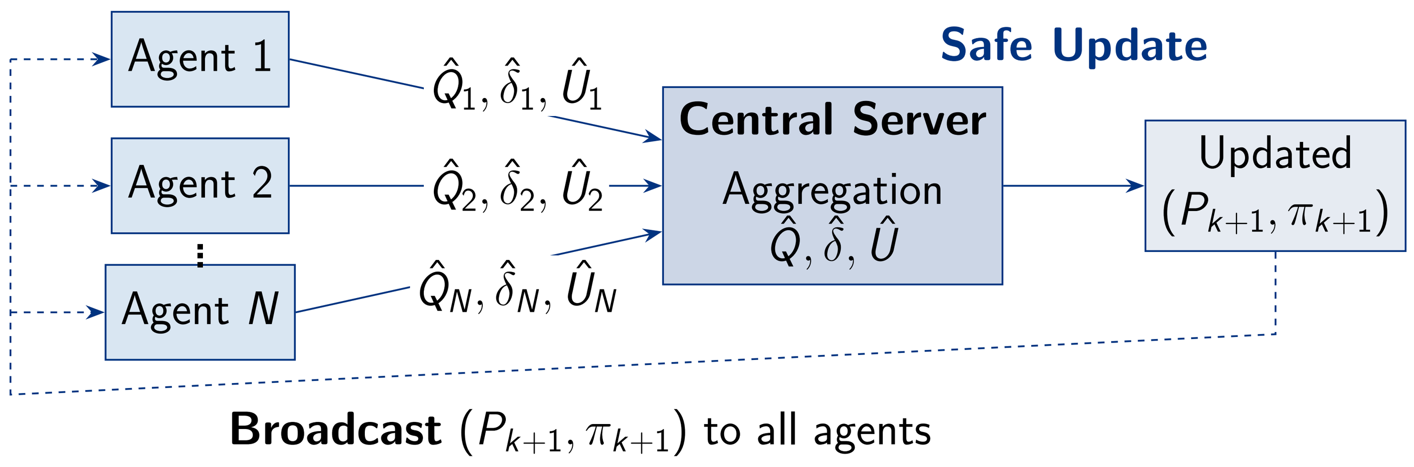
Our Solution: Federated Learning

We propose **F-SPMI** (Federated SPMI):

- Multiple agents explore in parallel
- Agents share learned value functions
- Central server aggregates knowledge
- **Safety guarantees preserved**

Faster learning through collaboration while maintaining safety!

F-SPMI Architecture



Local Statistics (per agent j):

- $\hat{Q}_j^{P_k,\pi_k}$: Q-function via Monte Carlo
- $\hat{\delta}_{\mu,j}^{P_k,\pi_k}$: γ-discounted state-action distribution
- $\hat{U}_j^{P_k,\pi_k}$: U-function estimate

GP-Enhanced Model Selection

Replace greedy model choice with **Gaussian Process UCB**:

$$P_{k+1} = \arg\max_{P \in \mathcal{P}} [\mu(P) + \beta_k \sigma(P)]$$

Balances *exploitation* (high predicted advantage μ) with *exploration* (high uncertainty σ) for principled model space search.

Safe Update Mechanism

Weighted Federated Averaging

The server aggregates local estimates:

$$\hat{Q}^{P_k,\pi_k}(s,a) = \frac{\sum_{j=1}^N M_j(s,a)\hat{Q}_j^{P_k,\pi_k}(s,a)}{\sum_{j=1}^N M_j(s,a)}$$

where $M_j(s,a)$ is the visit count by agent j.

Decoupled Bound (Theorem 3.3, Metelli et al.)

Performance improvement is lower bounded:

$$J_\mu^{P',\pi'} - J_\mu^{P,\pi} \geq \underbrace{\frac{A_{P,\pi,\mu}^{P',\pi} + A_{P,\pi,\mu}^{P,\pi'}}{1-\gamma}}_{\text{advantage}} - \underbrace{\frac{\gamma \Delta Q^{P,\pi} D}{2(1-\gamma)^2}}_{\text{penalization}} = \hat{B}(P',\pi')$$

Safe Step Sizes

Optimal update coefficients:

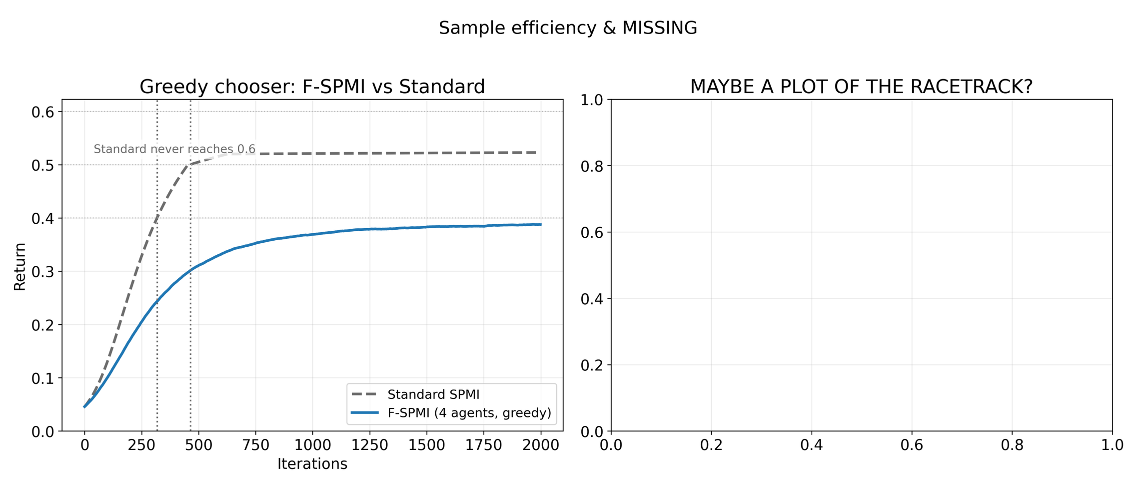
$$\alpha^*, \beta^* = \arg\max_{\alpha,\beta} \hat{B}(\alpha,\beta)$$

ensuring **monotonic performance improvement**.

Experimental Setup & Sample Efficiency

Domain: Racetrack-T1 Conf-MDP.
Metric: Discounted performance $J_\mu^{P,\pi} = \mu^\top V^{P,\pi}$ via exact evaluator; MC shown only as a diagnostic.

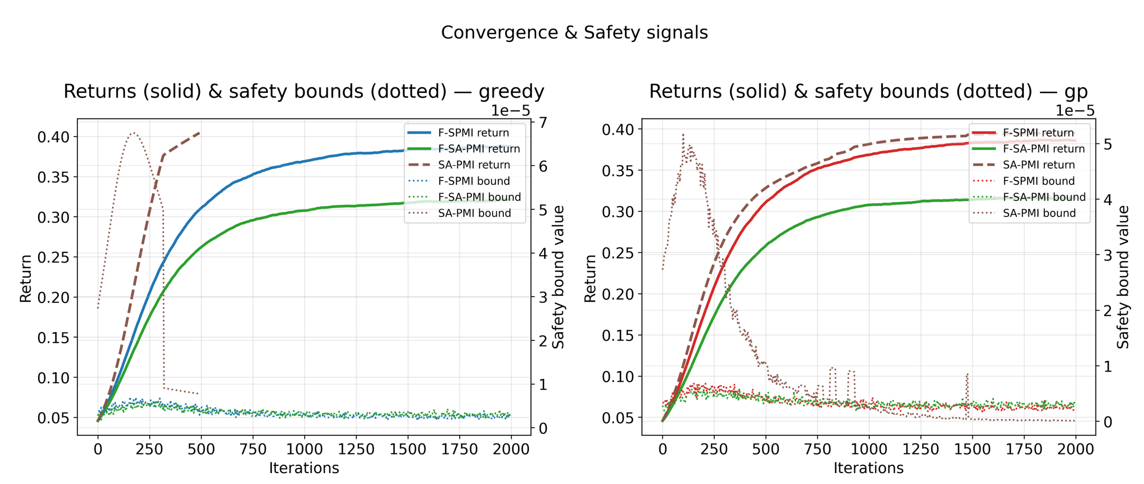
Observation: Standard SPMI reaches $J \geq 0.4$ at 534 iters and 0.5 at 1490; federated runs plateau ≈ 0.39 and never hit 0.4 within 2000 iters.



Convergence & Safety Guarantees

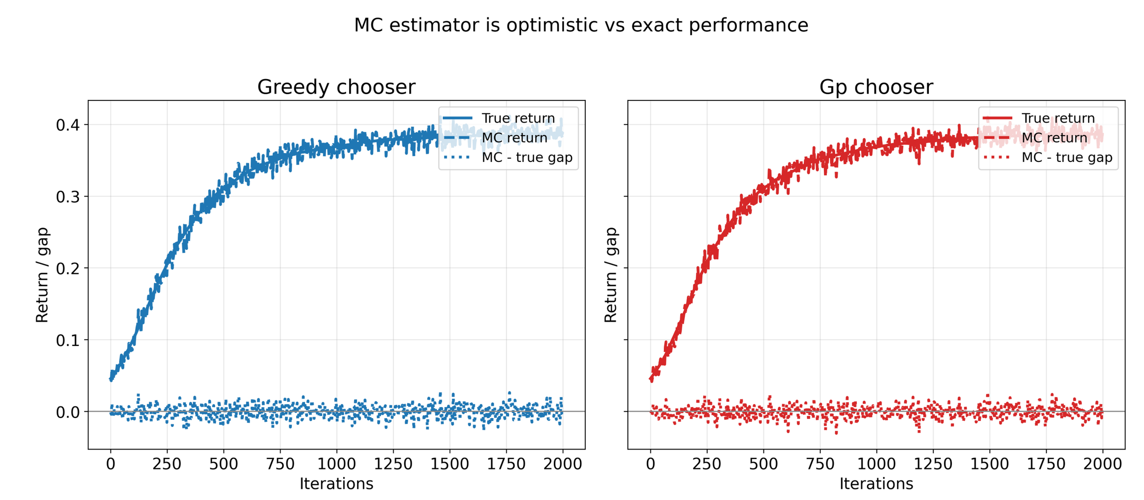
Safety signal: Dotted curves depict the theoretical bound $\hat{B}$; all methods keep it positive and shrinking ($\sim 10^{-6}$ by the end), indicating conservative updates.

Conclusion: On this track, federation does not reach the single-agent ceiling and the GP model chooser achieves comparable performance to the greedy case.



MC vs. True Performance

Alignment: MC return estimates with mean gap $< 10^{-3}$ to exact evaluator, thus MC tracks true with negligible bias.



Summary

Key Contributions:

- Federated SPMI executes **policy-model updates in parallel** while preserving safe improvement guarantees
- **Decoupled bound** yields monotonic policy/model refinement from finite-sample statistics
- Aggregation is **robust to stochasticity/heterogeneity**; MC returns track exact performance
- Optional **GP-UCB** target selection exploits structure of the model space

Limitations

Current Constraints:

- Assumes a **known finite model set** $\mathcal{P}$ and tabular state-action space
- **Synchronous** rounds; all agents must participate each update
- Communication cost scales with number of agents and statistics exchanged
- Safety guarantees impose limitations on performance an convergence

Outlook

Future Directions:

- Learn or approximate $\mathcal{P}$ from data; extend to latent/continuous model parametrizations
- **Function approximation** and policy-gradient variants for large/continuous spaces with safe steps
- **Asynchronous** or partial-participation protocols robust to malicious agents
- Deployment on robotic/autonomous platforms under bandwidth and safety constraints

Reference: Metelli et al., "Configurable Markov Decision Processes", ICML 2018